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Interbody fusion cages

Abstract

An implant for interbody fusion of vertebrae comprising a unibody cage structure having an enveloping cage volume and a minimized material volume. The cage structure comprises a first and a second generally planar ring member, each ring member formed from an opposing pair of lengthwise joists and an opposing pair of cross joists, the joists together forming a large opening through the ring member. The ring members are fixedly sandwiched on a plurality of support members, the support members holding the ring members in a spaced apart relationship to thereby provide a large void volume relative to the enveloping cage volume, to thereby allow for receipt of a large volume of bone graft within the cage structure.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority to the following applications, each of which is incorporated by reference herein in its entirety: the present application is a continuation of U.S. patent application Ser. No. 18/617,464 filed Mar. 26, 2024, which is pending; a continuation of U.S. patent application Ser. No. 17/203,225 filed Mar. 16, 2021, which issued as U.S. Pat. No. 11,938,038 on Mar. 26, 2024; U.S. patent application Ser. No. 17/202,833 filed Mar. 16, 2021, which is pending and which issues as U.S. Pat. No. 11,974,928 on May 7, 2024; U.S. patent application Ser. No. 16/304,430 filed Nov. 26, 2018, which issued as U.S. Pat. No. 10,952,869 on Mar. 23, 2021; U.S. patent application Ser. No. 16/304,430 claims the benefit of PCT/US2017/034883 filed May 27, 2017, which is expired; PCT/US2017/034883 claims the benefit of U.S. Provisional Patent Application 61/342,572 filed May 27, 2016, which is expired.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH

(1) Not applicable

FIELD OF THE INVENTION

(2) The present invention relates to spinal implants, and more particularly to lumbar and anterior cervical discectomy fusion cages configured to receive bone graft.

BACKGROUND OF THE INVENTION

(3) Various implants have been developed for use in lumbar interbody fusion (LIF) and anterior cervical discectomy fusion (ACDF), including cages. Current LIF and ACDF cages emphasize structural support over openness, which decreases options for the use of interbody graft to enhance fixation and fusion. There is a need for interbody fusion cages that overcome the shortcomings of the prior art. The present invention is directed primarily to improved fusion cage implants for transforaminal lumbar interbody fusion (TLIF), direct lateral lumbar interbody fusion (DLIF), anterior lumbar interbody fusion (ALIF), and anterior cervical discectomy fusion (ACDF), together with methods of use and of manufacture of such implants.

OBJECTS AND SUMMARY OF THE INVENTION

(4) It is an object of the inventions to provide improved spinal implants for use in lumbar interbody fusion ("LIF").

(5) It is an object of the inventions to provide LIF cages that provide sufficient structural support for LIF procedures while also featuring large internal void volumes communicating with large external openings, the larger volumes and openings relative to structure providing increased options for using interbody graft to enhance fixation and fusion.

(6) One object of the inventions is to provide a TLIF cage that can be readily inserted transforaminally and which provides a large void volume and external openings for use with interbody graft.

(7) Another object of the inventions is to provide a DLIF cage that can be affixed to selected lateral portions of the lumbar region.

(8) Another object of the inventions is to provide an ALIF cage that includes a removable clip for containing graft.

(9) Another object of the inventions is to provide anterior cervical discectomy fusion cages having the characteristics described herein.

(10) The foregoing objectives are achieved by providing LIF and cervical cages having the features described herein.

(11) Embodiments of the invention include an implant device for interbody fusion of lumbar or cervical vertebrae comprising a unibody cage structure having an enveloping cage volume and a minimized material volume. The cage structure comprises a first and a second generally planar ring member, each ring member formed from an opposing pair of lengthwise joists and an opposing pair of cross joists, the joists together forming a large opening through the ring member. The ring members are fixedly sandwiched on a plurality of support members, the support members holding the ring members in a spaced apart relationship to thereby provide a large void volume relative to the enveloping cage volume, to thereby allow for receipt of a large volume of bone graft within the cage structure.

(12) A ratio of the void volume to the enveloping cage volume may be between about 0.7 to about 0.9. The ratio of the void volume to the enveloping cage volume may be between about 0.79 to about 0.85. A ratio of the void volume to the material volume may be between about 3.8 to about 6.0.

(13) In embodiments, the ring members are positioned on opposing lateral sides of the cage, such that one of the lengthwise joists of each ring member serves as a superior loading surface, the other of the lengthwise joists of each ring member serves as an inferior loading surface, and the cross joists are configured to distribute load between opposing lengthwise joists. The cross joists may be curved, such that each ring member has a substantially race-track configuration. The cage may further comprise at least one substantially vertical strut sandwiched between the opposing lengthwise joists of each ring member to further distribute load. The at least one substantially vertical strut may comprise a pair of substantially vertical struts.

(14) In embodiments, the support members may comprise a base member on a posterior end of the ring members and an arrangement of arcuate struts on an anterior end of the ring members. The arrangement of arcuate struts may have an X-shape.

(15) In embodiments, the device is configured for use as a transforaminal lumbar interbody fusion (TLIF) cage. In TLIF embodiments, the ratio of the void volume to the enveloping cage volume may be between about 0.79 to about 0.81. A ratio of the void volume to the material volume is between about 3.9 to about 4.1. The joists may be between about 1.5 to about 2.5 mm in diameter. Each of the vertical struts may have an outer diameter of between about 1.5 mm to about 2.5 mm.

(16) In other embodiments, the first ring member is on a superior side of the cage and serves as a superior loading surface, the second ring member is on an inferior side of the cage and serves as an inferior loading surface, and the support members are a plurality of substantially vertical struts positioned and configured to distribute load between the ring members. The plurality of substantially vertical struts may include four corner struts, with each of the corner struts positioned substantially at an intersection between the lengthwise joists and the cross joists. A pair of the corner struts may be taller than an opposing pair of the corner struts, to thereby provide the cage structure with a lordosis angle. The lordosis angle may be between about 6 and about 20 degrees, such as between about 8 to about 12 degrees. At least one of the lengthwise joists may be arcuate. The first and second ring members may be substantially mirror images of one another.

(17) In embodiments, the device may be sized and configured for use as a direct lateral lumbar interbody fusion (DLIF) cage. The plurality of substantially vertical struts may further include a pair of internal struts, each of the internal struts substantially centered between the lengthwise joists on opposing sides of the cage to thereby distribute load between the lengthwise joists. In DLIF embodiments, the ratio of the void volume to the enveloping cage volume may be between about 0.84 to about 0.86. A ratio of the void volume to the material volume may be between about 5.6 to about 5.8. The joists may be about 4 mm wide and 1.5 mm thick. Each of the vertical struts may have an outer diameter of about 4 mm.

(18) In embodiments, the device is sized and configured for use as an anterior lumbar interbody fusion (ALIF) cage. In ALIF embodiments, the ratio of the void volume to the enveloping cage volume may be between about 0.83 to about 0.85. A ratio of the void volume to the material volume may be between about 5.1 to about 5.2. The may be about 4 mm wide and about 1.5 mm thick. The vertical struts may have an outer diameter of about 4 mm. A clip-on side cap member may be provided, the side cap member configured to removably attach to the vertical strut members to thereby assist in containing bone graft within the cage.

(19) In embodiments, the device is sized and configured for use as an anterior cervical discectomy fusion (ACDF) cage. In ACDF embodiments, the ratio of the void volume to the enveloping cage volume may be between about 0.81 to about 0.83. A ratio of the void volume to the material volume may be between about 4.68 to about 4.70. The joists are about 2.5 mm wide and about 0.75 mm thick. The vertical struts may have an outer diameter of about 2.5 mm.

(20) Methods of fusing adjacent posterior and anterior lumbar or cervical vertebrae of a patient are provided, such as providing an implant device as described herein, inserting the implant between adjacent vertebrae of said patient, and inserting a bone graft in the void volume of the implant.

(21) The foregoing and other objects, features, aspects and advantages of the invention will become more apparent from the following detailed description of the invention when considered in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a side view of one embodiment of a TLIF cage of the invention.

- (2) FIG. 2 is a top view of one embodiment of a TLIF cage of the invention.
- (3) FIG. 3 is a rear view of one embodiment of a TLIF cage of the invention.
- (4) FIG. 4 is a front view of one embodiment of a TLIF cage of the invention.
- (5) FIG. 5 is a front-side isometric view of one embodiment of a TLIF cage of the invention.
- (6) FIG. 6 is a rear-side isometric view of one embodiment of a TLIF cage of the invention.
- (7) FIG. 7 is a top side isometric view of one embodiment of a DLIF cage of the invention showing a rotational plate in an attachment orientation.
- (8) FIG. 8 is a rear view of one embodiment of a DLIF cage of the invention showing a rotational plate in an insertion orientation.
- (9) FIG. 9 is a side view of one embodiment of a DLIF cage of the invention showing a rotational plate in an attachment orientation.
- (10) FIG. 10 is a top view of one embodiment of a DLIF cage of the invention showing a rotation plate in an attachment orientation.
- (11) FIG. 11 is a rear view of one embodiment of a DLIF cage of the invention showing a rotational plate in an insertion orientation.
- (12) FIG. 12 is a side view of one embodiment of a DLIF cage of the invention showing a rotational plate in an insertion orientation.
- (13) FIG. 13 is a top view of one embodiment of a DLIF cage of the invention showing a rotational plate in an insertion orientation.
- (14) FIG. 14 is a rear-side isometric view of one embodiment of an ALIF cage of the invention.
- (15) FIG. 15 is a front-side isometric view of one embodiment of an ALIF cage of the invention.
- (16) FIG. 16 is a front view of one embodiment of an ALIF cage of the invention.
- (17) FIG. 17 is a top view of one embodiment of an ALIF cage of the invention.
- (18) FIG. 18 is a side view of one embodiment of an ALIF cage of the invention.
- (19) FIG. 19 is a rear-side isometric view of one embodiment of an ALIF cage of the invention.
- (20) FIG. 20 is an isometric view of one embodiment of a clip for an ALIF cage of the invention.
- (21) FIG. 21 is a top view of one embodiment of a clip for an ALIF cage of the invention.
- (22) FIG. 22 is an outside view of one embodiment of a clip for an ALIF cage of the invention.
- (23) FIG. 23 is an inside view of one embodiment of a clip for an ALIF cage of the invention.
- (24) FIG. 24 is a rear-side isometric view of one embodiment of an anterior cervical discectomy fusion cage of the invention.
- (25) FIG. 25 is a front-side isometric view of one embodiment of an anterior cervical discectomy fusion cage of the invention.
- (26) FIG. 26 is a front view of one embodiment of an anterior cervical discectomy fusion cage of the invention.
- (27) FIG. 27 is a top view of one embodiment of an anterior cervical discectomy fusion cage of the invention.
- (28) FIG. 28 is a side view of one embodiment of an anterior cervical discectomy fusion cage of the invention.
- (29) FIG. 29 is a side view of one embodiment of a TLIF cage of the invention.
- (30) FIG. 30 is a side view of one embodiment of a TLIF cage of the invention.
- (31) FIG. 31 is a rear view of one embodiment of a TLIF cage of the invention.
- (32) FIG. 32 is a front view of one embodiment of a TLIF cage of the invention.
- (33) FIG. 33 is a front-side isometric view of one embodiment of a TLIF cage of the invention.
- (34) FIG. 34 is a rear-side isometric view of one embodiment of a TLIF cage of the invention.
- (35) FIG. 35 is a side view of one embodiment of a DLIF cage of the invention.
- (36) FIG. 36 is a top-front isometric view of one embodiment of a DLIF cage of the invention.
- (37) FIG. 37 is a side isometric view of one embodiment of a DLIF cage of the invention.
- (38) FIG. 38 is a front view of one embodiment of a DLIF cage of the invention.
- (39) FIG. 39 is a top view of one embodiment of a DLIF cage of the invention.

(40) The inventions will be defined partly with respect to certain volumes and ratios. As used for the various cage embodiments described herein, “enveloping volume” means the volume of a rectangular box tightly surrounding the outermost edges of the cage. “Net enveloping volume” means “enveloping volume” minus the triangular wedge area that is not occupied by the cage due to the presence of a lordosis angle (one side higher than the other). As such, enveloping volume has a slightly large volume than net enveloping volume. “Material volume” (sometimes referred to as “structural volume”) means the volume occupied by the physical structure of the cages (struts, joists, etc.). “Void volume” means enveloping volume minus material volume. “Net void volume” means net enveloping volume minus material volume. Thus, it will be appreciated that enveloping volume includes both the material volume and the void volume, while net enveloping volume includes both the material volume and the net void volume. From these volumetric aspects, certain ratios can be determined; these ratios are important to the function and concept of the inventions described herein, as will be described below.

(41) TLIF Cage

(42) FIGS. 29-34 show an embodiment of a TLIF cage **1**. As shown in the view of FIG. 29, the TLIF cage **1** generally comprises a unibody frame structure having a posterior base **10**, an anterior end **40**, a plurality of lengthwise joist structures **20** extending between the posterior base **10** and the anterior end **40**, and an open area or void volume **50** formed internal to the posterior base **10**, the joist structures **20**, and the anterior end **40**.

(43) In the side view of FIG. 29, it can be seen that joist structures **20** are provided by a pair of opposing unibody load-bearing ring members **21**. Each ring member **21** is elongated into an oval or racetrack shape having opposing lengthwise portions **20**. As indicated in FIG. 30, each joist structure **20** includes a posterior support section **22**, an anterior support section **26**, and a lengthwise joist section **24** formed between the posterior and anterior support sections **22**, **26**. Ends of the posterior and anterior support sections **22**, **26** merge into a curved or arcuate configuration to thereby form the ring members **21**. The unibody, opposingly curved configuration of the ring members **21** helps support the load of opposing vertebrae on the lengthwise joist sections **24**. As can be observed in FIGS. 29-34, the ring members **21** are preferably annular in cross-section, such that each ring member **21** has a tubular appearance, to further increase the strength of the ring member **21** and particularly the joist sections **24** of the ring members **21**.

(44) To further strengthen the lengthwise joist sections **24**, one or more struts **60** may be provided between the lengthwise joist sections **24**. In the embodiment of FIGS. 29-34, a pair of struts **60** is provided on each ring member **21** between each pair of lengthwise joist sections **24**. For further strength, the ends each strut **60** can be formed as a curved or arcuate section **62** that merges into the adjacent joist **24**.

(45) As can be seen in FIGS. 30 and 32, the anterior support sections **26** of the ring members **21** are fixed in a spaced apart arrangement via a plurality of struts **46** at the anterior end **40** of the cage **1**. The anterior end **40** of the cage **1** is formed by a merger of the anterior struts **46**. When viewed from the lateral view of FIG. 30 or the superior/inferior view of FIG. 33, the opposing struts **46** form a generally U-shaped configuration, which further contributes to the strength of the cage **1**.

(46) As shown in FIG. 30, the posterior support sections **22** of the ring structures **21** are contiguously formed with the posterior base **10**, to provide a unibody structure. In this manner, the posterior base **10** serves, along with the struts **46** described above, to hold the opposing ring structures **21** in a rigid spaced arrangement. In one preferred embodiment shown in FIG. 2, the posterior support sections **22** extend from approximately a middle or mid-portion of the posterior base **10**. The features of the posterior base **10** are similar to those described above with reference to FIGS. 1-6, including an internally threaded screw hole **12**. However, it will be noted that the posterior base **10** of the embodiment of FIGS. 29-34 can be made shorter, due to the additional strength features, thus freeing up some space for greater void volume.

(47) In one embodiment, each of the TLIF cage **1** is preferably about 25 mm in length, from the

posterior base **10** to the anterior end **40**. The width of the cage **1** is about 9.5 mm, such as about 8.5 mm to about 10.5 mm. The height of the cage **1** is between about 7 to about 14 mm, such as about 10 mm. The joists of the ring members **21** have a major diameter of about 2.5 mm and a minor diameter about 1.5. The major diameter is preferably centered on the joist **24** in order to maximize loading bearing.

(48) As may be appreciated from the foregoing description, some of the improved characteristics of the TLIF cage **1** include a minimized volume, a reduced cross-sectional area yet with support strength comparable to a conventional TLIF cage, and a maximized internal void volume **50**. The void volume **50** enables use of higher amounts of bone graft in the internal void volume **50**. In embodiments, the overall or enveloping volume of the TLIF cage **1** is between about 2000 mm.sup.3 to about 2800 mm.sup.3, such as about 2375 mm.sup.3. The volume of the TLIF cage structure, or material volume, is between about 400 mm.sup.3 to 600 mm.sup.3, such as about 481 mm.sup.3; and the TLIF void volume **50** is between about 1,700 mm.sup.3 to about 2,100 mm.sup.3, such as about 1,894 mm.sup.3. The ratio of the void volume **50** to the overall cage volume may be between about 0.75 and about 0.85, such as about 0.80, or 80%. The ratio of the void volume **50** to the material volume of the TLIF cage structure is preferably between about 3.7 to about 4.3, such as about 4.0. These void volume ratios are much higher than the ratios found in conventional TLIF cages, and provide the advantages discussed herein.

(49) FIGS. **1-6** show an alternative embodiment of a TLIF cage **1**. As shown in the oblique view of FIG. **6**, the TLIF cage **1** generally comprises a unibody frame structure having a posterior base **10**, an anterior end **40**, a plurality of joist structures **20** extending between the posterior base **10** and the anterior end **40**, and an open area or void volume **50** formed internal to the posterior base **10**, the joist structures **20**, and the anterior end **40**.

(50) As shown in FIG. **1**, the posterior base **10** is preferably cube shaped, although other shapes could be used. The sides of the cube are preferably about 7 mm in length. As shown in FIG. **3**, a screw hole **12** is formed in the posterior base **10**. The screw hole **12** is preferably about 3.5 mm in diameter. The screw hole **12** is internally threaded **30** for receipt of a fixation screw or a threaded end of an insertion device. The internal thread **30** may be a machine thread. In one embodiment, the insertion device is a lengthwise device (not shown) having a handle on one end and a cage engagement end on an opposing end, the cage engagement end featuring two or more tines spaced to engage matching apertures on the posterior base **10**, the tines serving to stabilize the insertion device on the cage **1** during insertion of a screw through the cage engagement end and into the screw hole **12** to thereby removably secure the insertion device on the cage **1**.

(51) As indicated in FIG. **2**, each joist structure **20** includes a posterior support section **22**, an anterior support section **26**, and a lengthwise joist section **24** formed between the posterior and anterior support sections **22**, **26**. Each of the posterior and anterior support sections **22**, **26** has a curved or arcuate configuration to help support the load of opposing vertebrae on the lengthwise joist sections **24**.

(52) As can be seen in the posterior view of FIG. **3** and the anterior view of FIG. **4**, the four joist structures **20** can be characterized as axially arranged or aligned around a longitudinal axis of the cage **1**. As shown in FIGS. **1-3**, the posterior support sections **22** of the joist structures **20** extend from the posterior base **10**. The posterior support sections **22** are contiguously formed with the posterior base **10**, to provide a unibody structure. In one preferred embodiment shown in FIG. **2**, the posterior support sections **22** extend from approximately a middle or mid-portion of the posterior base **10**.

(53) As can be seen in FIG. **4**, the anterior support sections **26** of the joist structures merge at the anterior end **40** of the cage **1**. Thus, the anterior end **40** of the cage **1** is formed by a merger of anterior support sections **26**. When viewed from the lateral view of FIG. **1** or the superior/inferior view of FIG. **2**, the opposing anterior support sections **26** of the joist structures **20** form a generally U-shaped configuration, which contributes to the strength of the cage **1**.

(54) As indicated in FIGS. 1-2, some or all of the lengthwise sections **24** of each of the joist structures **20** preferably have a somewhat pancaked profile, with differing superior/inferior and lateral dimensions. When viewed superiorly or inferiorly, as shown in FIG. 2, the support side **24S** of the lengthwise section **24** has a wide dimension. This wide dimension of the support side **24S** allows for loading on a vertebral body surface and also provides greater surface area for bone fixation. However, when viewed laterally, as shown in FIG. 1, the lateral side **24L** of the lengthwise joist section **24** has a narrow dimension. The narrow dimension provides greater space for receipt of bone graft in the internal open area **50**.

(55) In one embodiment, each of the joist structures **20** is preferably about 24 mm in length, from the posterior base **10** to the anterior end **40**. The flattened or wide dimension of the support side **24S** of the lengthwise joist **24** is preferably between about 1.5 to 3 mm, and preferably 2.5 mm. The narrow dimension of the lateral side **24L** of the lengthwise joist **24** is preferably about 1 mm. Thus, in a preferred embodiment, the ratio of the support side **24S** to the lateral side **24L** is preferably between about 1.5:1 and about 3.0:1, or most preferably about 2.5:1.

(56) The height and width of the cage **1** is preferably about 10 mm. The length of the cage **1** is preferably about 24 mm from the posterior end **10** to the anterior end **40**.

(57) To accommodate patient sizes, an assortment of sizes of TLIF cages **1** can be provided. The height of the cages **1** can vary from 7 mm to 14 mm, for example in 1 mm increments. Two or more lengths can be provided, such as 22 mm and 24 mm.

(58) The base frame of the TLIF cage **1** is formed or made of metal to provide strength and rigidity to the cage **1**. The metal is preferably a body compatible metal, which will typically be titanium or cobalt chrome. The cage **1** can be used uncoated, in which case the surface is preferably laser sintered to facilitate in-growth of surrounding bone into the surface of the cage. In other embodiments, the cage **1** is preferably over-coated with polyether-ether-ketone (PEEK) **58** circumferentially. The internal rigid first layer of metal, which is typically laser sintered titanium, acts as an internal skeleton, providing strength for the overall device. The PEEK over-coating **58**, or second layer, makes the cage **1** partially or completely radiopaque. The second layer **58** functions to increase the surface area superior and inferior for loading against adjacent endplates, and also provides some elasticity to the TLIF structure **1**. The superior and inferior loading surfaces **24S** may be plasma coated with porous titanium **60** to form a third layer. In other embodiments, the entire PEEK layer is plasma coated with porous titanium. The porous titanium outer coating or layer **60** facilitates bone in-growth for the endplates of the vertebral bodies for improved fixation. The optional PEEK over-coating and titanium plasma coating increase the overall dimensions of the cage **1** by about 17 microns. Alternatively, the metal cage can be downsized so that the aforementioned over-coated embodiments have substantially the same dimensions a sintered titanium embodiment. In yet another embodiment, the cage can be coated with PEEK integrated with hydroxyapatite in lieu of an additional outside layer of titanium.

(59) In the TLIF cage **1** embodiment of FIGS. 1-6, the overall or enveloping volume of the TLIF cage (structure and void) is about 2400 mm.sup.3, the volume of the TLIF cage structure (joists and base) is about 583 mm.sup.3, and the TLIF void volume **50** is about 1817 mm.sup.3. The ratio of the void volume **50** to the enveloping cage volume is preferably between about 0.65 and 0.9, and most preferably about 0.75. The ratio of the void volume **50** to the volume of the TLIF cage structure is preferably between about 2.5 and 4.0, and most preferably about 3.0. These void volume ratios are much higher than the ratios found in conventional TLIF cages, and provide the advantages discussed herein.

(60) A method of manufacturing a composite TLIF device **1** includes first forming or assembling an internal structure or cage **1** from a metal, such as of titanium or cobalt chrome. A coating or layer of PEEK **58** is then applied over all or a substantial portion of the internal cage **1**. A coating of porous titanium **60** is then applied to the PEEK **58** second layer and is bonded to the second layer. The porous titanium **60** is applied to the superior and inferior loading surfaces **24S**, or optionally to all

or a substantial portion of the cage **1**. Bone graft can be packed into the internal void volume **50** of the TLIF cage **1**. Optionally, a fourth layer, comprising a bioabsorbable condom, can be placed over the TLIF cage to envelop the cage and retain the graft prior to final implantation. The graft packing and enveloping steps can be carried out intra-operatively by the surgeon.

(61) DLIF Cage

(62) In an exemplary embodiment of a DLIF cage **100** is shown in FIGS. **35-39**. As shown in FIG. **36**, the DLIF cage **100** comprises generally a unibody cage portion **102** formed from opposing ring members **170S**, **170I** and having a voluminous central opening **150**. The sides of the DLIF cage **100** have a plurality of large openings **123**, **125**, **127** formed between support columns **161**, **162**, **165** and leading into the voluminous void volume **150**. Together, the voluminous central opening **150** and side openings **123**, **125**, **127** contribute to and form a large void volume **103** configured for receipt of bone graft or other adjuncts.

(63) As indicated in FIG. **38**, the DLIF cage portion **102** can be considered to have a superior side surface **S** and an inferior side surface **I**. As indicated in FIG. **36**, the DLIF cage portion **102** can be considered to have a front or anterior side **A** and a back or posterior side **B**. Further, as indicated in FIG. **38**, the cage portion **102** can be considered to have a leading or insertion end **112** and a trailing end **110** on lateral sides thereof. A slight slope, for example 1 degree, can be provided between the insertion end **112** and the trailing end **110**.

(64) As indicated in FIG. **36**, a first or superior side of the cage **102** is formed from a generally planar ring member **170S**, while a second or inferior side of the cage **102** is formed from an opposing generally planar ring member **170I**. The opposing ring members **170S**, **170I** are maintained in spaced relationship by a plurality of strut members **161**, **162**, **165**. The strut members **161**, **162**, **165** may be integrally formed with the ring members **170S**, **170I**. Each ring member **170S**, **170I** is formed from a first or anterior lengthwise joist **120A** and a second or posterior lengthwise joist **120B**. The anterior and posterior joists **120A**, **120B** are interconnected by a pair of opposing lateral joists **124**, the joists **120A**, **120B**, **124** together forming an integral surface **171**. As shown in FIG. **39**, the various joists **120A**, **120B**, **124** are preferably joined at their respective ends. In the embodiment of FIGS. **35-39**, the inferior side ring member **170I** of the cage portion **102** presents substantially a mirror image of the superior ring member **170S**.

(65) As shown in FIG. **39**, the superior and inferior ring members **170S**, **170I** are interconnected by a plurality of struts **161**, **162**, **165**. In the embodiment shown, a pair of anterior struts **161A**, **162A** interconnect opposing ends of the anterior superior and inferior joists **120**. Likewise, a pair of posterior struts **162A**, **162B** interconnect opposing ends of the posterior superior and inferior joists **322S**, **322I**. A central pair of struts **165A**, **165B** is provided for distributing vertical load. For maximum strength, the struts **161**, **162**, **165** are sandwiched between and integrally formed with the superior and interior sides **170S**, **170I**.

(66) In the embodiment of FIG. **39**, each of the first and second lengthwise joist portions **120A**, **120B** is bowed outward, such that each lengthwise joist **120A**, **120B** is arcuate. As indicated in FIG. **39**, this arrangement enables a large central cavity or void volume **150** within the cage portion **102**. Each of the lengthwise joist portions **120** may be centrally bowed away from the centerline of the cage **100** and peaked at the center strut **165A**, **165B** so as to strengthen the joists, provide additional strength, and increase the open area.

(67) The sides formed by the spaced lengthwise joist portions **120A**, **120B** and cross joists **124** preferably have at least one side opening **123** passing therethrough. As seen in the anterior/posterior side view of FIG. **38**, each of the joists **120** preferably has a first or lateral side opening **123** and a second or medial side opening **125**. As indicated in FIG. **38**, each side opening **123**, **125** passes entirely through a sidewall of the joist **120**, such that the void volume **150** communicates with the exterior through the openings **123**, **125**. This configuration facilitates release of bone graft from the void volume **150** into the vertebral space adjacent the cage **100**. The side openings **123**, **125** are sized and configured such that the strength of the joist portions **120** and

cage structure **102** not significantly compromised while maximizing the potential for release of graft through the openings **123, 125**.

(68) Preferred dimensions of the DLIF cage of FIGS. **35-39** will now be described. It is expected that due to various anatomical sizes in the patient population, various sizes of DLIF cages **100** according to the invention will be provided, such as in an implant kit or surgical tray. Thus, variations from these dimensions may be made without deviating from the spirit and scope of the invention. The length of the cage portion **102**, measured from the first lateral side **110** to the second lateral side **112**, is preferably 50 mm. The width of the cage portion **102**, measured from the outer edge of the peaks of the first and second opposing lengthwise joists **120A, 120B** is preferably 25 mm. The height of the cage portion **102**, is preferably between about 9 to 10 mm. In sloped embodiments, the cage portion may have a slope of about 8 degrees. The height of the cage portion **102** on the leading side may be about 9 mm (e.g. 9.1 mm), while the height on the trailing side may be about 10 mm. The joists **120, 124** are preferably about 4 mm wide and about 2 mm thick. The support struts **161, 162, 165** may have a major diameter of about 4 mm.

(69) To accommodate patient sizes, an assortment of sizes of DLIF cages **100** can be provided. The height of the cages **100** can vary from about 8 mm to 16 mm, for example in 1 mm increments. Two, three or more lengths can be provided, such as 48, 50 and 52 mm. Two, three or more widths can also be provided, such as 23, 24 and 25 mm. Various lordosis angles can be provided, such as 8 degrees.

(70) In a preferred embodiment, the enveloping volume of the DLIF cage is between about 11,500 mm.³ to about 13,500 mm.³, such as about 12,500 mm.³. The net enveloping volume is between about 11,000 mm.³ to 13,000 mm.³, such as about 11,938 mm.³.

(71) The material volume of the DLIF cage structure is between about 1800 mm.³ to about 2000 mm.³, such as about 1868 mm.³. The void volume is between about 9,500 mm.³ to about 11,500 mm.³, such as about 10,632 mm.³. The net void volume is between about 9,000 mm.³ to about 11,000 mm.³, such as about 10,069 mm.³.

(72) The ratio of the void volume to the enveloping volume is preferably between about 0.75 to about 0.9, such as about 0.85. The ratio of the net void volume to the net enveloping volume is preferably about 0.75 to about 0.9, such as about 0.84.

(73) The ratio of the void volume to the material volume of the DLIF cage structure is preferably between about 5.0 to about 6.5, such as about 5.69. The ratio of the net void volume to the material volume is between about 4.8 to about 6.3, such as about 5.39. These void volume ratios are much higher than the ratios found in conventional DLIF cages, and provide the advantages discussed herein.

(74) DLIF Cage with Rotational Plate

(75) FIGS. **7-13** show views of one embodiment of applicant's DLIF cage having a rotational plate **100**. As shown in the perspective view of FIG. **7**, the DLIF cage **100** comprises generally a unibody cage portion **102** having a voluminous central opening or void volume **150** configured for receipt of bone graft or other adjuncts and a rotational plate **180** rotatably disposed on a first lateral side **110** of the cage portion **102**.

(76) As shown in FIG. **8**, the rotational plate **180** is provided with first and second opposing screw bores **183, 185** configured for receipt of spinal bone screws. The screw bores **183, 185** will typically be unthreaded, so as to allow the surgeon to select the screw insertion angle, which allows for polyaxial fixation.

(77) FIGS. **11-13** show the rotational plate **180** in an insertion configuration in which the plate portion **180** is rotated to or aligned with the first lateral side **110** at about zero degrees, in a horizontal plane orientation. In contrast, FIGS. **7-10** show the rotational plate **180** in a fixation configuration in which the plate **180** has been rotated 90 degrees, into a vertical plane orientation. In this configuration, the surgeon can selectively secure the DLIF cage **100** to the lumbar vertebrae by passing screws through one or both of the first and second screw bores **183, 185**. In addition to

the zero and 90 degree positions shown in the drawings, the surgeon has the option of rotating the plate to any desired position relative to the patient's internal anatomy. Thus, the DLIF cage with rotational plate **100** is configured for insert into the lumbar space and to allow for selection of and fixation to a preferred location of anatomical bone. This selective capability is particularly useful when operating on patients who have soft or otherwise compromised bone, where it may be difficult or impossible to adequately secure a plate in certain regions of the lumbar vertebrae.

(78) As can be seen in FIG. **10**, the cage portion **102** includes a first lateral side **110**, a second or opposing lateral side **112**, and first and second lengthwise strut portions **120A**, **120B** extending between the first and second lateral sides **110**, **112**. In the embodiment of FIG. **10**, each of the first and second lengthwise strut portions **120A**, **120B** is bowed outward, such that each lengthwise strut **120A**, **120B** is arcuate. As indicated in FIG. **10**, this arrangement enables a large central cavity or void volume **150** within the cage portion **102**.

(79) At least one of the lengthwise strut portions **120** preferably has at least one side opening **123** passing therethrough. As seen in the anterior/posterior side view of FIG. **9**, each of the struts **120** preferably has a first or lateral side opening **123** and a second or medial side opening **125**. As indicated in FIGS. **9** and **12**, each side opening **123**, **125** passes entirely through a sidewall of the strut **120**, such that the void volume **150** communicates with the exterior through the openings **123**, **125**. This configuration facilitates release of bone graft from the void volume **150** into the vertebral space adjacent the cage **100**. The side openings **123**, **125** are sized and configured such that the strength of the strut portions **120** and cage structure **102** not significantly compromised while maximizing the potential for release of graft through the openings **123**, **125**.

(80) As can be seen in the embodiment of FIG. **12**, a plurality of loading surface designs **140** are provided or formed on both the superior **114** and inferior **116** sides of the cage portion **102**. In the embodiment shown in the drawings, the loading surface designs **140** are ridges **140** arranged in parallel rows aligned along the anterior-posterior dimension of the cage **102**. However, other configurations could be used for the loading surface designs **140**, such as cross-hatches, broken rows, or medial-lateral rows. Alternatively, the bone engagement surface of the DLIF cage **100** can be smooth (not shown). In all surface embodiments, porous titanium can be used for bone ingrowth.

(81) The DLIF cage portion **102** is preferably formed of PEEK material, but it can be made of other biocompatible metals, such as titanium or cobalt chrome.

(82) In an alternative embodiment, the rotational plate **180** is provided on a separate clip (not shown). The separate clip is preferably made of PEEK and the rotational plate is made of metal. The separate clip is configured to clip onto lateral struts, such as a pair of lateral struts, formed on the DLIF cage **100**. Alternatively, the separate clip and the rotational plate can be made of PEEK, or both can be made of metal.

(83) A shaving instrument (not shown) can be provided for preparing the lateral aspect of the vertebral body to allow the plate to rotate freely relative to the vertebrae prior to screw placement. The instrument will progressively clear a space, such as an annular space, in the bone adjacent the implant site to allow the plate to rotate freely until a final position for the plate is selected.

(84) ALIF Cage with Side Clip

(85) FIGS. **14-23** show views of embodiments of applicant's ALIF cage **300**. As shown in the perspective view of FIG. **14**, the ALIF cage of the invention **300** generally comprises a cage portion **302** having a voluminous interior cavity or void volume **350**, large side openings **348A**, **348P**, **348L**, and snap-on/clip-on anterior and posterior side caps **380A**, **380P** for use in selectively sealing the anterior and posterior side openings **348A**, **348P**. The cage portion **302** is formed from opposing superior and inferior joist portions **322**, **323**, **324**, **325** which are interconnected and spaced apart by a plurality of support struts **310**, as will be described below.

(86) As indicated in FIG. **16**, the cage portion **302** has a superior side surface **321S** and an inferior side surface **321I**. As indicated in FIG. **17**, the cage portion has an anterior side **303A** and a

posterior side **303P**. As shown in the top or superior view of FIG. **17**, the superior side surface **321S** is formed from a superior anterior joist **322S** and an opposing superior posterior joist **323S**. The superior anterior and posterior joists **322S**, **323S** are interconnected by a pair of opposing superior lateral joists **324S**, **325S**, the joists together forming an integral superior surface **321S**. As shown in FIG. **17**, the various joists **322S**, **323S**, **324S**, **325S** are preferably joined at their respective ends. The inferior side surface **321I** of the cage portion **302** preferably presents substantially a mirror image of the superior side **321S**, with an inferior anterior joist **322I** and an inferior posterior joists **323I** interconnected by a pair of inferior lateral joists **324I**, **325I**. As shown in FIG. **19**, the superior and inferior sides **321S**, **321I** are interconnected by a plurality of struts **310**. In the embodiment shown, a pair of anterior struts **310A** interconnect opposing ends of the anterior superior and inferior joists **321S**, **321I**. Likewise, a pair of posterior struts **310P** interconnect opposing ends of the posterior superior and inferior joists **322S**, **322I**. For maximum strength, the struts **310A**, **310P** are sandwiched between and integrally formed with the superior and inferior sides **321S**, **321I**.

(87) The ALIF cage portion **302** includes various features that facilitate an ALIF procedure. As can be seen in the top view of FIG. **17**, the anterior joist **322** is bowed outwards anteriorly in an arcuate manner. The posterior joist **323** is bowed outwards posteriorly, but in a more flattened or gentle curve than the anterior joist **322**. The lateral joists **324**, **325** are preferably straight. As indicated in FIG. **18**, an anterior-to-posterior lordosis slope is preferably built into the cage portion **302**. The lordosis slope is preferably about 6-10 degrees, and most preferably about 8 degrees. Larger slopes can be used to provide greater lordosis. The slope can be built into both sides of the cage **302**, in which case the slope will be about 4 degrees per side, or built into only one side of the cage **302**, such that one side is flat and the other side has a slope of about 8 degrees. This slope can be facilitated by making the anterior struts **310A** longer than the posterior struts **310P**. In one preferred embodiment, the anterior struts **310A** are 9 mm in height while the posterior struts **310P** are 8 mm in height.

(88) The described cage structure **302** provides a relatively large void volume **350** relative to the volume of the cage structure **302**, as well as large superior and inferior openings and large anterior, posterior and opposing side openings **348A**, **348P**, **348L**. This allows for the use of a large amount of bone graft in the void volume **350**, as well as release of the graft through the openings. Use of the void volume **350** and openings is further facilitated by selective use of the ALIF clips **380**, as described below. However, despite the large void volume **350** and openings, the structure of the cage **302**, as described herein, provides the cage **302** with sufficient strength for use as an ALIF implant.

(89) The ALIF cage portion **302** is formed from a bio-compatible metal such as titanium or cobalt chrome. At least a portion of the outer surface of the ALIF cage portion **302** is laser-sintered with a plasma titanium powder coating **60** for optimal bony in-growth and secure fixation against vertebral endplates. In one embodiment, the entire surface of each of the loading superior and inferior outer surfaces **321S**, **321I** are plasma coated with titanium **60**.

(90) Details of embodiments of the anterior and posterior side caps **380A**, **380P** will now be discussed. As shown in FIGS. **14-15**, the anterior and posterior side caps **380A**, **380P** are configured to snap-on to the struts **310** of the cage **302**. In an embodiment shown in FIGS. **20-23**, each side cap **380** comprises a unibody structure including a pair of wings **391**, **392**. Opposing ends of each wing **391**, **392** are provided with an annular wall **387** formed on the inner surface **381** of the ALIF cap **380**. When viewed from an exterior or outer side **382**, a bend **383** is formed in the juncture where the annular wall **387** begins, as shown in FIG. **22**. Each annular wall **387** encloses a respective inner cavity **386**. A portion of the annular wall **387** extends beyond 180 degrees, leaving an opening for receipt of a strut **310** of the ALIF cage **302** into the cavity **386**, in a snap-fit engagement. By “snap-fit engagement,” it will be understood that annular wall **387** is sufficiently resilient to deflect or compress elastically so as to store elastic energy when a strut **310** is pushed into cavity **386**. The annular wall **387** has an inner diameter that is complementary to an outer

diameter of the strut **310**, but slightly larger in size, so as to secure the cap **380** on the struts **310** while minimizing motion between the cap **380** and the ALIF cage **302**.

(91) To further enhance the ALIF cage construct **300**, one or both of the caps **380** is preferably provided with an inner strut **390**, as shown in FIGS. **20-23**. The inner strut **390** is sized and configured to provide additional loading support between the inferior and superior joists **319**, **320** when attached to the ALIF cage **302**. As shown in FIG. **21**, the inner strut **390** is preferably formed on the inner surface **381** of the cap **380**, and is preferably centered or substantially centered on the cap **380**, between the opposing wings **391**, **392**. The inner strut **390** preferably has a rounded peak and a rounded or sloped interface with the inner surface **381** of the cap **380**. As can be appreciated from the construct views of FIGS. **14-15**, when the cap **380** is attached to the cage **302**, the height of the inner strut **390** is dimensioned so as to abut against the inferior surface **327** of the superior joists **322S**, **323S** and the superior surface **328** of the inferior joists **322I**, **323S**. In this manner, the inner struts **390** provide load support between the joists **322**, **323**, and particularly between the central portions of the joists, where compressive forces from the vertebrae receive less countervailing support from the cage struts **310**. Thus, the large openings **348A**, **348P** on the anterior and posterior sides of the cage **302** facilitate the use of interbody graft, while the removable caps **380** selectively enclose the openings **348A**, **348P** while providing supplemental loading support akin to that provided by the cage struts **310**.

(92) The posterior and anterior side caps **380A**, **380P** are preferably made of PEEK. Other materials can be used, provided that such materials are biocompatible and enable the side caps **380A**, **380P** to selectively snap on to or off of the struts **310** in a sufficiently secure engagement.

(93) Preferred dimensions of the ALIF cage **302** and clips **380** will now be described. It is expected that due to variations in anatomical size in the patient population, various sizes of ALIF cage constructs **300** according to the invention will be provided, such as in an implant kit or surgical tray. Thus, it will be appreciated that variations from these dimensions may be made without deviating from the spirit and scope of the invention.

(94) In one preferred embodiment, the ALIF cage **302** of the invention has a length of 37 mm, a depth of 27 mm, and a maximum height of 10 mm. The joists **322**, **323**, **324**, **325** have a height of only 1.5 mm when viewed from the posterior or anterior side, as indicated in FIG. **16**. The joists **322**, **323**, **324**, **325** have a width of 4 mm when viewed from the top or bottom, as indicated in FIG. **17**.

(95) The struts **310** have a diameter of 4 mm. The anterior struts **310** have a height of 10 mm, while the posterior struts **310** have a height of 9 mm. This provides the aforementioned anterior-to-posterior slope of about 8 degrees.

(96) The ALIF clip **380** has a length of 37 mm, or approximately the length of the ALIF cage **302**. The center strut **390** has a depth or height of 4.5 mm. The wings **391**, **392** have a depth or height of 4.5 mm and are 0.75 mm thick. The height of the anterior ALIF clip **380** is 7 mm, in order to substantially fill the anterior opening **348A**. Likewise, the height of the posterior ALIF clip **380** is 6 mm, which substantially fills the posterior opening **348P**.

(97) To accommodate patient sizes, an assortment of sizes of ALIF cages **300** can be provided. The height of the cages **300** can vary from about 8 mm to 16 mm, for example in 1 mm increments. Two, three or more lengths can be provided, such as 35, 37 and 39 mm. Two, three or more widths or depths can also be provided, such as 26, 27 and 28 mm. Various lordosis angles can be provided.

(98) In a preferred embodiment, the overall volume of the ALIF cage (structure and void) is about 9900 mm.^{sup.3}, the volume of the ALIF cage structure (joists and struts) is about 1821 mm.^{sup.3}, and the ALIF void volume **350** is about 8,169 mm.^{sup.3}. The ratio of the void volume **350** to the overall cage volume is preferably between about 0.7 and 0.9, and most preferably about 0.8. The ratio of the void volume **350** to the volume of the ALIF cage structure is preferably between about 4.2 and 4.7, and most preferably about 4.5. These void volume ratios are much higher than the ratios found in conventional ALIF cages, and provide the advantages discussed herein.

(99) In one exemplary embodiment, the ALIF cage structure has a width of about 25 mm, a length of about 35 mm, a minimum height of 6.5 mm, a maximum height of 9.8 mm, and a lordosis angle of 8 degrees. The joists have a width of 4 mm and a thickness of 1.5 mm.

(100) In an exemplary embodiment, the enveloping volume of the ALIF cage is between about 7,500 mm.³ to about 9,500 mm.³, such as about 8,575 mm.³. The net enveloping volume is between about 6,000 mm.³ to about 8,000 mm.³, such as about 7,131 mm.³.

(101) The material volume of the ALIF cage structure is between about 1,100 mm.³ to about 1,700 mm.³, such as about 1,398 mm.³. The void volume is between about 6,000 mm.³ to about 8,000 mm.³, such as about 7176 mm.³. The net void volume is between about 4500 mm.³ to about 7,000 mm.³, such as about 5732 mm.³.

(102) The ratio of the void volume to the enveloping volume is preferably between about 0.7 to about 0.9, such as about 0.84. The ratio of the net void volume to the net enveloping volume is preferably about 0.7 to about 0.9, such as about 0.80.

(103) The ratio of the void volume to the material volume of the ALIF cage structure is preferably between about 4.6 to about 5.6, such as about 5.13. The ratio of the net void volume to the material volume is between about 3.6 to about 4.6, such as about 4.1. These void volume ratios are much higher than the ratios found in conventional ALIF cages, and provide the advantages discussed herein.

(104) An important objective of the implants of the invention is to provide a greater volume of bone graft as well as optimized communication with the surrounding disk space area. The cages allow for a high volumetric expansion of bone graft. For example, the TLIF cage of the invention allows for the use of about 6-8 times more bone graft than a conventional TLIF cage, such as TLIF cages commercially available from Medtronic. The cages of the invention also allow the graft material to fully communicate with the disk space while maintaining support of the disc space.

(105) For each of the cages of the invention, the minimal frame structures allow, in addition to the use of greater amounts of graft material, a minimal radiographic foot print compared to conventional LIF or ACDF cages. This characteristic allows for greater visualization when viewing and monitoring the post-operative progression of interbody fusion.

(106) For each of the cages of the invention, the titanium surface is preferably configured to allow for bone in-growth on some or all of the outer surfaces of the cage. Sintering can be used to achieve an appropriate in-growth surface, in a manner known in the art. Alternatively, in some embodiments, the cage can be coated with PEEK integrated with hydroxyapatite in lieu of an additional outside layer of titanium.

(107) When PEEK is applied to one of the frames described herein, it can be applied in various ways. One method is to spray PEEK onto the metal frame of the cage. The sprayed PEEK is preferably implant grade PEEK. Although spraying of industrial grade PEEK to form a coating has been used outside of the medical implant field, applicants are unaware of the previous use of spraying of implantable PEEK onto implants. Another method is to dip or submerge the cage into a solution of PEEK and then remove the cage from the PEEK solution. Brushing or other methods can be used, provided that the PEEK sufficiently covers and adheres to the underlying frame.

(108) One of the advantages of the invention is that it works with commercially available bone graft materials while enhancing the ability of such bone grafts to fuse vertebrae. Bone grafts promote bone healing through an osteogenic, osteoconductive or osteoinductive mechanism. Examples of grafts that could be used with the cages of the invention include autografts and allografts. The graft can include natural or recombinant BMPs or bone marrow aspirate. The graft can be in the form of putties, pastes or granules, or combinations thereof, and can include antibiotic or other anti-bacterial agents. A commercial brand that could be used with the cages of the invention is INFUSE® bone graft from Medtronic.

(109) Anterior Cervical Discectomy Fusion Cage

(110) FIGS. 24-28 show views of embodiments of applicant's anterior cervical discectomy fusion

(ACDF) cage **1300**. The ACDF cage **1300** incorporates features of the ALIF cage as well as the general characteristics and advantages described herein with respect to lumbar interbody fusion cages.

(111) As shown in the perspective view of FIG. **24**, the ALIF cage of the invention **300** generally comprises a cage portion **1302** having a voluminous interior cavity or void volume **1350**, and large side openings **1348A**, **1348P**, **1348L**. The cage portion **1302** is formed from opposing superior and inferior joist portions **1322**, **1323**, **1324**, **1325** which are interconnected and spaced apart by a plurality of support struts **1310**, as will be described below.

(112) As indicated in FIG. **26**, the cage portion **1302** has a superior side surface **1321S** and an inferior side surface **1321I**. As indicated in FIG. **27**, the cage portion **1302** has an anterior side **1303A** and a posterior side **1303P**. As shown in the top or superior view of FIG. **27**, the superior side surface **1321S** is formed from a superior anterior joist **1322S** and an opposing superior posterior joist **1323S**. The superior anterior and posterior joists **1322S**, **1323S** are interconnected by a pair of opposing superior lateral joists **1324S**, **1325S**, the joists together forming an integral superior surface **1321S**. As shown in FIG. **27**, the various joists **1322S**, **1323S**, **1324S**, **1325S** are preferably joined at their respective ends. The inferior side surface **1321I** of the cage portion **1302** preferably presents substantially a mirror image of the superior side **1321S**, with an inferior anterior joist **1322I** and an inferior posterior joist **1323I** interconnected by a pair of inferior lateral joists **1324I**, **1325I**. As shown in FIG. **24**, the superior and inferior sides **1321S**, **1321I** are interconnected by a plurality of struts **1310**. In the embodiment shown, a pair of anterior struts **1310A** interconnect opposing ends of the anterior superior and inferior joists **1321S**, **1321I**. Likewise, a pair of posterior struts **1310P** interconnect opposing ends of the posterior superior and inferior joists **1322S**, **1322I**. For maximum strength, the struts **1310A**, **1310P** are sandwiched between and integrally formed with the superior and inferior sides **1321S**, **1321I**.

(113) The ACDF cage portion **1302** includes various features that facilitate a ACDF procedure. As can be seen in the top view of FIG. **27**, the anterior joist **1322** is bowed outwards anteriorly in an arcuate manner. The posterior joist **1323** is bowed outwards posteriorly in an arcuate manner. The lateral joists **1324**, **1325** are preferably straight. As indicated in FIG. **28**, an anterior-to-posterior slope is preferably built into the cage portion **1302**. The slope is preferably about 7 to 9 degrees, and most preferably about 8 degrees. The slope can be built into both sides of the cage **1302**, in which case the slope will be about 4 degrees per side, or built into only one side of the cage **1302**, such that one side is flat and the other side has a slope of about 8 degrees. This slope can be facilitated by making the anterior struts **1310A** longer than the posterior struts **1310P**. In one embodiment, the anterior struts **1310A** are about 4.8 mm in height while the posterior struts **1310P** are about 3.9 mm in height.

(114) The described cage structure **1302** provides a relatively large void volume **1350** relative to the volume of the cage structure **1302**, as well as large superior and inferior openings and large anterior, posterior and opposing side openings **1348A**, **1348P**, **1348L**. This allows for the use of a large amount of bone graft in the void volume **1350**, as well as release of the graft through the openings. However, despite the large void volume **1350** and openings, the structure of the cage **1302**, as described herein, provides the cage **1302** with sufficient strength for use as a ACDF implant.

(115) The ACDF cage portion **1302** is formed from a bio-compatible metal such as titanium or cobalt chrome. At least a portion of the outer surface of the ACDF cage portion **1302** is laser-sintered with a plasma titanium powder coating **60** for optimal bony in-growth and secure fixation against vertebral endplates. In one embodiment, the entire surface of each of the loading superior and inferior outer surfaces **1321S**, **1321I** are plasma coated with titanium **60**.

(116) Preferred dimensions of the ACDF cage **1302** will now be described. It is expected that due to variations in anatomical size in the patient population, various sizes of ACDF cage constructs **1300** according to the invention will be provided, such as in an implant kit or surgical tray. Thus, it

will be appreciated that variations from these dimensions may be made without deviating from the spirit and scope of the invention.

(117) To accommodate patient sizes, an assortment of sizes of ACDF cages **1300** can be provided. The height of the ACDF cages **1300** can vary from about 5 mm to 10 mm, for example in 1 mm increments. Two, three or more lengths can be provided, such as 12, 14 or 16 mm. Two, three or more widths or depths can also be provided, such as 14 and 16 mm.

(118) In one exemplary embodiment, the ACDF cage structure has a width of about 14 mm, a length of about 16 mm, a minimum height of 4.4 mm, a maximum height of 6.3 mm, and a lordosis angle of 8 degrees. The joists have a width of 2.5 mm and a thickness of 0.75 mm. The struts **1310** have a diameter of 2.5 mm.

(119) In an exemplary embodiment, the enveloping volume of the ACDF cage is between about 1,100 mm³ to about 1,700 mm³, such as about 1,411 mm³. The net enveloping volume is between about 900 mm³ to 1,500 mm³, such as about 1,198 mm³.

(120) The material volume of the ACDF cage structure is between about 200 mm³ to about 300 mm³, such as about 248 mm³. The void volume is between about 900 mm³ to about 1,400 mm³, such as about 1,163 mm³. The net void volume is between about 700 mm³ to about 1200 mm³, such as about 950 mm³.

(121) The ratio of the void volume to the enveloping volume is preferably between about 0.7 to about 0.9, such as about 0.82. The ratio of the net void volume to the net enveloping volume is preferably about 0.73 to about 0.87, such as about 0.79 (e.g. 950 mm³:1198 mm³).

(122) The ratio of the void volume to the material volume of the ACDF cage structure is preferably between about 4.2 to about 5.2, such as about 4.69 (e.g. 1,163 mm³:248 mm³). The ratio of the net void volume to the material volume is between about 3.3 to about 4.3, such as about 3.83 (e.g. 950 mm³:248 mm³). These void volume ratios are much higher than the ratios found in conventional ACDF cages, and provide the advantages discussed herein.

(123) Although the present invention has been described in terms of specific embodiments, it is anticipated that alterations and modifications thereof will no doubt become apparent to those skilled in the art. It is therefore intended that the following claims be interpreted as covering all alterations and modifications that fall within the true spirit and scope of the invention.

Claims

1. An implant device for interbody fusion of lumbar or cervical vertebrae comprising: a unibody cage structure, the cage structure comprising titanium, the cage structure having an enveloping cage volume and a minimized material volume, the cage structure comprising a first and a second generally planar ring member, each ring member formed from an opposing pair of lengthwise joists and an opposing pair of cross joists, the joists together forming a large opening through the ring member, the ring members fixedly sandwiched on a plurality of support members, the support members holding the ring members in a spaced apart relationship to thereby provide a large void volume relative to the enveloping cage volume, to thereby allow for receipt of a large volume of bone graft within the cage structure, wherein a ratio of the void volume to the enveloping cage volume is between about 0.7 to about 0.9.

2. The device of claim 1, wherein the ratio of the void volume to the enveloping cage volume is between about 0.79 to about 0.85.

3. The device of claim 1, wherein a ratio of the void volume to the material volume is between about 3.8 to about 6.0.

4. The device of claim 1, wherein the first ring member is on a superior side of the cage and serves as a superior loading surface, the second ring member is on an inferior side of the cage and serves as an inferior loading surface, and the support members are a plurality of substantially vertical struts positioned and configured to distribute load between the ring members.

5. The device of claim 4, wherein the plurality of substantially vertical struts includes four corner struts, and each of the corner struts is positioned substantially at an intersection between the lengthwise joists and the cross joists.
 6. The device of claim 5, wherein a pair of the corner struts is taller than an opposing pair of the corner struts, to thereby provide the cage structure with a lordosis angle.
 7. The device of claim 6, wherein the lordosis angle is between about 6 and about 20 degrees.
 8. The device of claim 7, wherein the lordosis angle is between about 8 to about 12 degrees.
 9. The device of claim 6, wherein the device is sized and configured for use as a direct lateral lumbar interbody fusion (DLIF) cage.
 10. The device of claim 9, wherein the ratio of the void volume to the enveloping cage volume is between about 0.84 to about 0.86.
 11. The device of claim 9, wherein a ratio of the void volume to the material volume is between about 5.6 to about 5.8.
 12. The device of claim 9, wherein the joists are about 4 mm wide.
 13. The device of claim 9, wherein the joists are about 1.5 mm thick.
 14. The device of claim 9, wherein each of the vertical struts has an outer diameter of about 4 mm.
 15. The device of claim 6, wherein the device is sized and configured for use as an anterior lumbar interbody fusion (ALIF) cage.
 16. The device of claim 15, wherein the ratio of the void volume to the enveloping cage volume is between about 0.83 to about 0.85.
 17. The device of claim 15, wherein a ratio of the void volume to the material volume is between about 5.1 to about 5.2.
 18. The device of claim 15, wherein the joists are about 4 mm wide.
 19. The device of claim 15, wherein the joists are about 1.5 mm thick.
 20. The device of claim 15, wherein each of the vertical struts has an outer diameter of about 4 mm.
 21. The device of claim 6, wherein the device is sized and configured for use as an anterior cervical discectomy fusion (ACDF) cage.
 22. The device of claim 21, wherein the ratio of the void volume to the enveloping cage volume is between about 0.81 to about 0.83.
 23. The device of claim 21, wherein a ratio of the void volume to the material volume is between about 4.68 to about 4.70.
 24. The device of claim 21, wherein the joists are about 2.5 mm wide.
 25. The device of claim 21, wherein the joists are about 0.75 mm thick.
 26. The device of claim 21, wherein each of the vertical struts has an outer diameter of about 2.5 mm.
 27. The device of claim 1, wherein a surface of the cage structure is laser sintered to facilitate in-growth of bone into the surface of the cage structure.
 28. The device of claim 4, wherein at least one of the lengthwise joists is arcuate.
 29. The device of claim 4, wherein the first and second ring members are substantially mirror images of one another.
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